# Kushal Singh Kushwaha

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## Projects

**Local RAG Pipeline** | GitHub | *FastAPI, React/TypeScript, SQLite, sentence-transformers, Ollama* Jul 2026

- Built a local-first RAG system enabling grounded question-answering over uploaded PDFs using Qwen 2.5, with zero external API calls or data leaving the local machine
- Implemented a full retrieval pipeline - chunking, sentence-transformer embeddings, cosine similarity search - ingesting a 471-page PDF ( 610KB) in  55s, with embeddings persisted in SQLite
- Designed a streaming FastAPI backend using Server-Sent Events, delivering full grounded responses (retrieval through token-by-token generation) over a 471-page document in  18s end-to-end

**AceStar Nexus** | GitHub | *Tauri (Rust), FastAPI, React/TypeScript, SQLite, PyInstaller* Jun 2026

- Built a persistent always-on-top desktop companion widget branched from Nexus, shipped as a standalone Windows executable ( 31MB total) via PyInstaller sidecar — no terminal required,  5s cold start
- Implemented a priority-focused UI showing one urgent task at a time with live metric snapshots, expanding into full goal, reminder, internship, and fitness management
- Integrated MCP servers for LeetCode, GitHub, and fitness data with native desktop notifications via tauri-plugin-notification, running at  208MB total idle memory (126MB webview + 72MB backend + 10MB app)

**Nexus** | GitHub | *FastAPI, React/TypeScript, MCP, SQLite, Ollama* May 2026

- Built an open-source local-first developer dashboard aggregating live GitHub, LeetCode, and fitness data via custom MCP servers into a unified FastAPI backend
- Designed a modular MCP server architecture enabling independent data source integration with a shared React/TypeScript frontend
- Integrated optional Ollama-powered AI brief summarizing daily developer metrics across 3 custom MCP servers (GitHub, LeetCode, fitness) with  4s response time and zero data leaving the local machine

**Cipher-Vault** | GitHub | *Python, Cryptography, AES-256-GCM, Client-Side Encryption* Jan 2025

- Built a web-based secure storage tool implementing AES-256-GCM encryption entirely client-side; server stores only ciphertext with zero access to plaintext data
- Designed a server-blind architecture where even a fully compromised server exposes no user data
- Implemented clean separation between encryption logic and file I/O, making the cryptographic core independently reusable

## Technical Skills

**Languages**: Python, C++, C, JavaScript, TypeScript
**Frameworks & Libraries**: FastAPI, React, Tauri, Node.js (familiar), Ollama, SQLModel
**Backend & Systems**: REST APIs, MCP protocol, JWT authentication, WebSocket communication, AES-256-GCM encryption, local system automation
**Developer Tools**: Git, GitHub, VS Code, SQLite, PyInstaller, Linux (basic)
**Problem Solving**: Leetcode 100+ problems solved (Neetcode 150, C++)

## Education

**Amity University Lucknow** Lucknow, Uttar Pradesh, India
*B.Tech Computer Science and Engineering — CGPA: 9.43* *2024 – 2028*